CSE-1102

CHAPTER-03

CONSTANT, VARIABLES & DATA TYPES

06-08-2026

C Token

i) keyword (int, if, return)
ii) identifier (age, sum, main)
iii) constant (10, 3.14, 'A'
iv) special symbol (;, (), {})
v) Operators (+, -, *, =, %)
vi) strings

i) keyword: some words which is reserved for C-language is called keyword.

Example: int, char, float, if, else, while, return.

Rules for keywords

i) keywords must be written in lowercase.
ii) keywords are fixed
iii) basic building blocks for program statement.
iv) must not contain white space

ii) Identifiers: Identifiers refer to the name of variables, functions and arrays. These are user-defined names and consist of a sequence of letters & digits.

[Program এর variable, function, array ইত্যাদির নাম করণের জন্য]

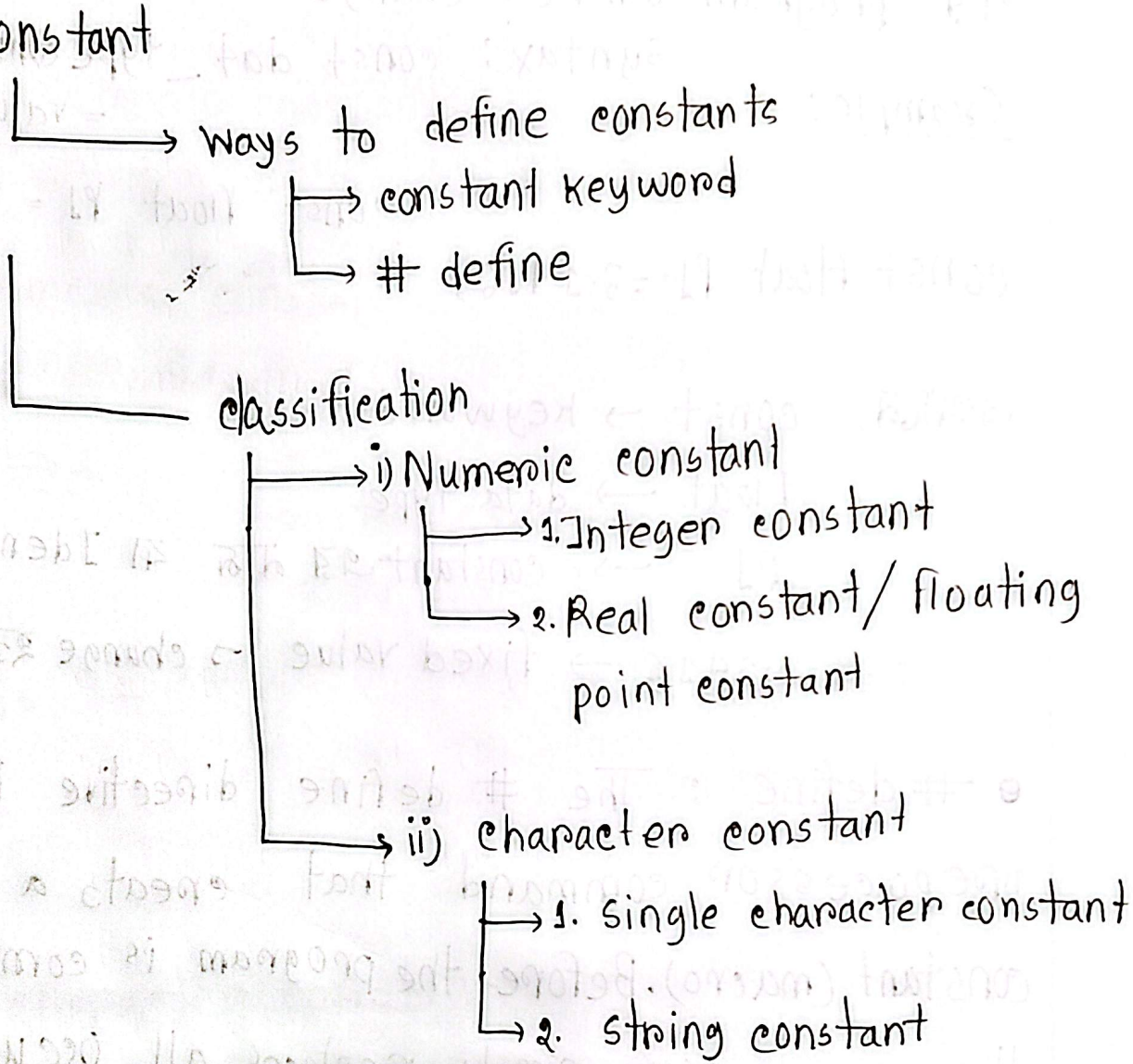
Example : int age → age হিসেবে, এটা Identifier.

age, sum, number, main, multiplication etc.

Rules for Identifiers

1. First character must be an alphabet (or underscore)
2. Must consist of only letters, digit or underscore.
3. Only first 31 characters are significant.
4. Cannot use a keyword.
5. Must not contain white space.

(iii) constant: constants in C refer to a fixed value that doesnot change during the execution of program.



- constant keywords: The constant keyword is a type qualifier that makes a variable read only after its initial assignment. This method offers type safety and follows normal variable scoping rules (local or global)

→ একটা value দিয়ে একবার কাজ করা যায়, প্রয়োজনে বার বার declare করে কাজ করতে হয়।

অর্থাৎ constant ব্যবহার করে কোনো variable-কে এমনভাবে declare করা যায় যে একবার value দেওয়ার পর program চলাকালে change করা যায় না।

Syntax: const dat_type constant_name =value

Example:

const float PI = 3.1416;

const float PI = 3.1416;

এখানে, const → keyword
float → data type
PI → constant-এর নাম বা Identifier
3.1416 → fixed value → change হয় না

- #define : The # define directive is a preprocessor command that creats a symbolic constant (macro). Before the program is compiled, the preprocessor simply replaces all occurrences of the symbolic name with its defined value in the source code.

• Syntax: # define CONSTANT NAME Value

Example: # define PI 3.1416

i) Numeric constants

সংখ্যা দিয়ে প্রকাশ করা হয় তাই Numeric constant

Numeric constant

- 1. Integer constant (Value পূর্ণসংখ্যা)
  (10, -5, 2)
- 2. Real constant / floating constant (ভগ্নাংশ থাকবে)
  (3.14, -0.5)

Example: float price = 25.50;

- float → data type
- price → Identifier
- 25.50 → floating constant

ii) character constant

- 1. Single character constant
  (একটি মাত্র character থাকবে) + single quatation

  Example: char grade = 'A';

- 2. String character constant
  (এক বা একাধিক character double quatation)

  Example: printf ("Hello");

iv) Special symbol / Punctuators

এগুলো program এর structure/syntax তৈরি করে

(; , , , ( ), { }, [ ])

v) Operators *

যেগুলো ব্যবহার করে Operation বা কাজ করা হয়

(+ - * / %) [আরও details ৭ Unit)

vi) String

Double quotation (" ") এর মধ্যে একাধিক character কে string বা string literal বলে।

Escape sequence

In the C programming language, an escape sequence is a sequence of characters used within a character or string literal to represent a character that is non-printable, has a special meaning, or can not typed directly from a keyboard. All escape sequences start with a backslash (\).

\n → print a new line

\t → print tab (বড় gap এর জন্য)

\\ → prints a single backlash (\ একটা print হয়)

(↑↓) \v → prints a vertical tab (vertical ekta gap)
space

\' → single quote > print হয়

\" → Double "

\b → backspace

Variable has some rules:

i) They must begin with a letter. Some system permit underscore as the first character.

ii) ANSI standard recognizes a length of 31 characters.

iii) Upercase and lowercase are significant.

iv) It show not be a keyword

v) White space is not allowed

al may be used
: constant
; the execution
y take different
ding execution.

or 32 bites
bites

32 bites

floating point
bites

- (ii) Derived
- (iii) Usero defined
  - 1. Arrays
  - 2. functions
  - 3. structures
  - 4. pointers
- 5. Void - The void types has no value

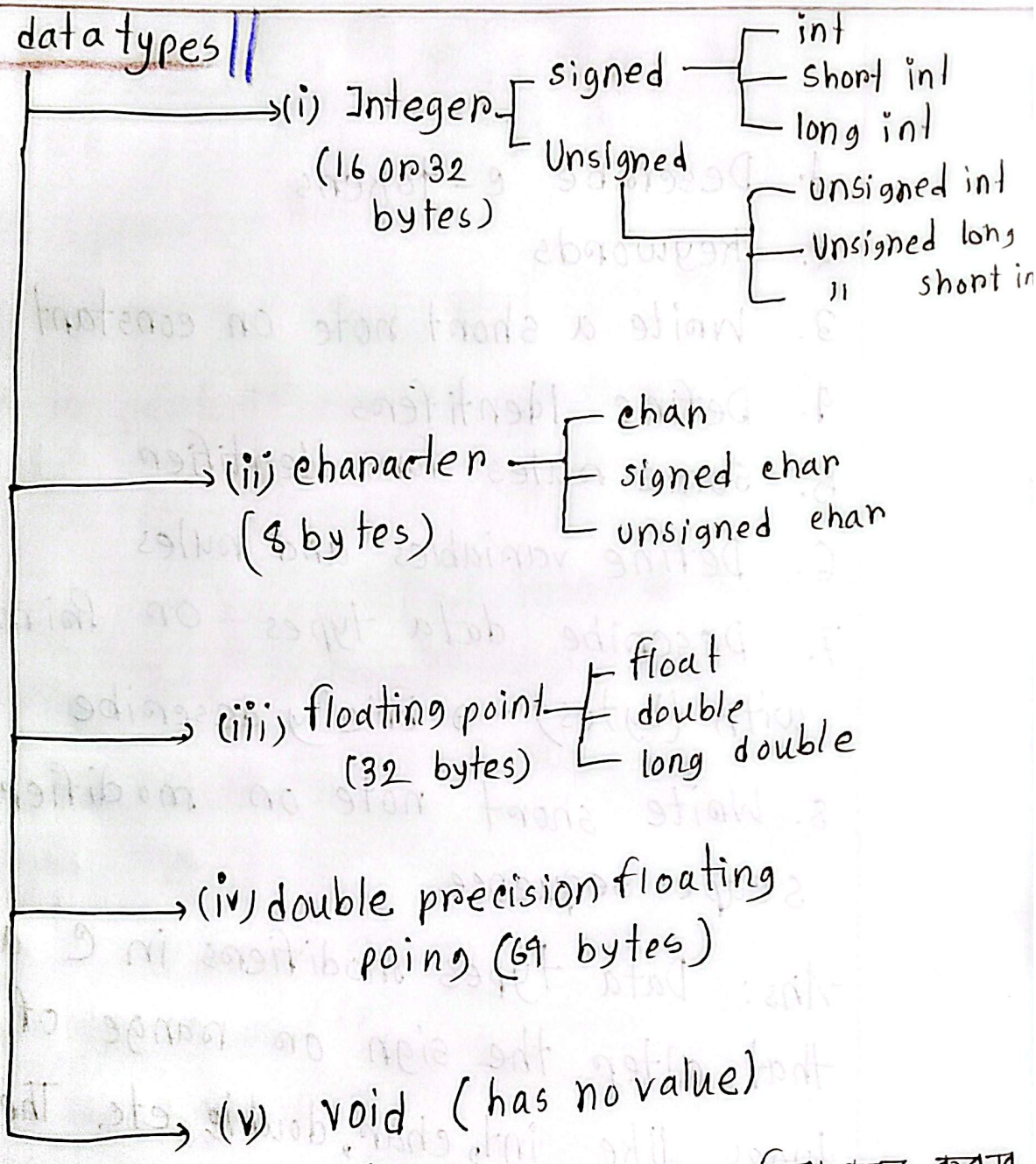
Variables

A variable is a data name that may be used to stroe a data value. Unlike constant that remain unchanged during the execution of a program, a variable may take different values at different times during execution.

Datatypes

- (i) Primary data types
  - 1. Integer – 16 or 32 bites
  - 2. Character – 8 bites
  - 3. Floating point – 32 bites
  - 4. Double precision floating point – 64 bites
  - 5. Void – The void types has no value
- (ii) Derived
- (iii) Usero defined
  - 1. Arrays
  - 2. functions
  - 3. structures
  - 4. pointers

Primary data types



↳ ইচ্ছামতো char নিয়ে কাজ করার জন্য ব্যবহার করা হয়।

signed, unsigned, long, short → 4 টা modifier

Unit 03

Some important Question

1. Describe e-tokens
2. keywords
3. Write a short note on constant
4. Define Identifiers
5. some rules for Identifier
6. Define variables and rules
7. Describe data types or Primarmy datatypes with (bytes) → briefly describe
8. Write short note on modifiers and scape sequence.

Ans: Data types modifiers in C are keywords that alter the sign or range of basic data types like int, char, double etc. There are 4 main modifiers:

- short
- long
- signed
- unsigned

CHAPTER- 04
OPERATORS & EXPRESSIONS

1. Define Operator?

ANS: An Operator is a symbol that tells the computer to perform certain mathematical and logical manupulation.

C Operators can be classified into a number catagories. They include:

1. Arithmetic Operators
2. Relational Operator
3. Logical Operator
4. Assignment Operator
5. Conditional Operator
6. Bitwise Operator
7. Increment and decrement Operator
8. Special Operator

1. Arithmetic Operator

Symbol	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Reminder - - - > ভাগশেষ আসবে

2. Relation Operator

Operator	Meaning & example
<	is less than
<=	is less than or equal to
>	is greater than
>=	is greater than or equal to
==	is equal to
!=	is not equal to

3. Logical Operator

Symbol	Name
&&	AND
||	OR
!	NOT

4. Assignment Operator

Assignment Operator	short hand Operator
a = a+1	a+=1
a = a-1	a-=1
a = a*(n-1)	a*=n+1
a = a/(n+1)	a/=n+1
a = a%b	a%=b

5. Conditional Operator: (? :)

Syntax : condition ? value 1 : value 2 ;

Example :

```
int a = 10 , b = 20;
int max = (a>b) ? a:b;
```

এখানে, condition a>b false তাই

max = 20

এটা অনেক অংশ ছোট if-else এর মতো কাজ করে।

6. Bitwise Operator: A bitwise Operator is used for manupulating of data at bit level.

Operator	meaning
&	Bitwise AND
|	Bitwise OR
^	Bitwise X-OR
<<	Shift left
>>	Shift right

(i) Bitwise AND: → (গুণ এর মতো)

```
  0000 1101 1
  0101 0101 0
-------------
  0000 0101 0
```

→ '0' থাকলেই '0'

(ii) Bitwise OR : (যোগ এর মতো)

```
  0000 1101 1
  0101 0101 0
-------------
  0101 1101 1
```

→ '1' থাকলেই '1'

iii) Bitwise X-OR:

```
  0000 1101 1
  0101 0101 0
-------------
  0101 1000 1
```

→ same হলে '0'
না হলে '1'

iv) shift Right :

212 → 11010100

212 >> 2 → 00110101

212 >> 7 → 00000001

v) Left shift:

212 → 11010100

212 << 2 → 1101010000

212 << 7 → 01000000

vii) 7) Increment and Decrement Operator

Increment	Decrement
The increment operator increases the value of a variable by 1	The decrement Operator decreases the value of a varible by 1
It has 2 types (i) pre-increament (++a) (ii) Post-increament (a++)	It has 2 types (i) pre-decrement (--a) (ii) post-decrement (a--)
Let m = 7 Pre increment: y = ++m y = 8 m = 8	Let m = 7 Pre decrement: y = --m y = 6 m = 6
Post increment: y = m++ y = 7 m = 8	Post decrement y = m-- y = 7 m = 6

8) Special Operator:

(sizeof, &, *, ,, ., ->)

Logical Operator দিয়ে (True/False)

ধরি, a = 10, b = 20, c = 5

Example 1 (a > b) && (b > c)

10 > 20 && 20 > 5

0 && 1

= 0 / False

Example 2 (a < b) || (a < c)

10 < 20 || 10 < 5

1 || 0

= 1 / True

Example 3 !(a > b)

!(10 > 20)

! 0 = 1 / True

Operator Precedence

Example 4 $(a<b \;\&\&\; b>c) \;||\; (a==c)$

$(10<20 \;\&\&\; 20>5) \;||\; 10==5$

$1 \;\&\&\; 1 \;||\; 0$

$= 1 \;||\; 0$

$= 1$ / True

Example 5 $!(a<b \;\&\&\; b<c) \;||\; (a>c)$

$a<b$

$10<20$

$= 1$

$!(1 \;\&\&\; 0) \;||\; 1$

$= !0 \;||\; 1$

$= 1 \;||\; 1$

$= 1$

$b<c$

$= 20<5$

$= 0$

$a>c$

$= 10>5$

$= 1$

NOT > AND > OR

( ) → ! * / % → + − → Relational (<, >) → Equality

NOT

&& AND

|| OR

Assignment

Example 1:

((10+5) * 2 > 20 && ! (8-3 == 5)) || (12/3 < 5)

= (15 * 2 > 20 && ! (8-3 == 5)) || (12/3 < 5)

= (15*2 > 20 && ! (5 == 5)) || (12/3 < 5)

= (15*2 > 20 && ! 1) || (12/3 < 5)

= (15*2 > 20 && ! 1) || (4 < 5)

= (30 > 20 && ! 1) || 1

= (1 && 0) || 1

= 0 || 1

= 1

Write down some rules for increment and decrement operator.

ANS.: Rules for increment and decrement operators:

1. Both are array operators that require a variable operand.

2. Postfix (++ or --) uses the original value of a++, a-- in the expression first, then increments the variable.

3. Prefix (++ or --) increments the variable ~~a+++~~ ++a, --a first then uses the ~~ta~~ new value in the expression.

4. The precedence and associativity are the same as array (+) and array (-) operators.

Briefly describe Operator presidence and associativity with example:

ANS: Operator precedence;

Operator precedence determine which Operator will perform first in the expression. more than one operator with different preecedence.

Operator Associativity:

Operator associativity is used when two Operator of same presedence appear in an expression. Associativity can be either right to left or left to right.

CamScanner